

HIERARCHICAL GENERALIZED LINEAR MODELS WITH R. APPLICATIONS IN PLANT/ANIMAL BREEDING AND GENETICS

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Recap

- The estimates for an ordinary least squares problem is $\hat{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$
- The hat matrix $\mathbf{H}$ “puts the hat” on $\mathbf{y}$, i.e. $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$
- The *hat values* are the diagonal elements of $\mathbf{H}$.
- The estimates for a weighted least squares problem is $\hat{\beta} = (\mathbf{X}^T \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{W} \mathbf{y}$
- Estimates for a generalized linear model are computed by iterating between weighted least squares

Course outline

- **Linear regression**
 - diagnostic plots
- **Weighted least squares**
- **Generalized linear models**
- **Linear mixed models**
 - Estimating fixed and random effects
 - It's a weighted least squares problem!
 - Estimating variance components
 - Application on repeated observations
 - Fitting an animal model using the additive relationship matrix
 - Smoothing

GLM

Poisson

$$L = \prod_{i=1}^n \frac{\mu^{y_i} e^{-\mu}}{y_i!}$$

$$\log(\mu) = X\beta$$

Binomial

$$L = \prod_{i=1}^n \binom{n}{y_i} \mu^{y_i} (1 - \mu)^{n - y_i}$$

$$\text{logit}(\mu) = X\beta$$

Gaussian

$$L = \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2\sigma^2}(y_i - \mu)^2}$$

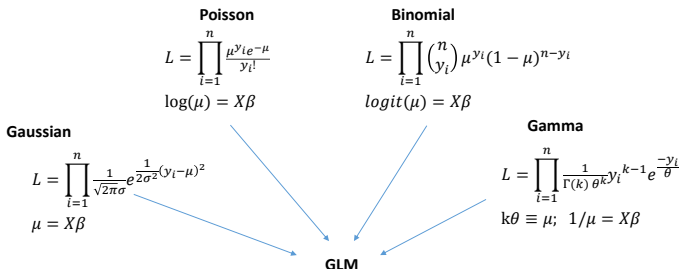
$$\mu = X\beta$$

Gamma

$$L = \prod_{i=1}^n \frac{1}{\Gamma(k) \theta^k} y_i^{k-1} e^{-\frac{y_i}{\theta}}$$

$$k\theta \equiv \mu; \quad 1/\mu = X\beta$$

GLM



Common estimation algorithm using iterative regression

- Fast and easy to implement
- ***Linear regression model checking tools!!!***

Why estimate the residual variance using a Gamma GLM?

- We get standard errors of the estimated value
- We can include linear predictors for the residual variance

Why does it work?

- If $X \sim N(0,1)$ then the distribution of X^2 is chi-squared distributed with 1 degree of freedom, i.e. χ_1^2 .
- χ_1^2 is a special case of the Gamma distribution.
- The standardized residuals from a linear model are normally distributed and their squared values are then Gamma distributed.
- Explained in more detail in *Ronnegard et al. 2010. Genetics Selection Evolution 42:8*.

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How to do it in R

Code that gives us the residual variance in two different ways.

```
y <- c(1.18, 3.28, 0.27, 3.69, 2.50, 5.53,  
       3.55, 3.24, 2.95, 5.29, 4.83, 3.94, 3.22)  
year <- c(2001, rep(2002,4), rep(2003,3), rep(2004,5))  
mod1 <- lm(y ~ year)  
summary(mod1)  
  
res1 <- residuals(mod1)  
hv1 <- hatvalues(mod1)  
response1 <- (res1^2)/(1-hv1)  
weight1 <- (1-hv1)/2  
glm1 <- glm( response1 ~ 1, weights = weight1 ,  
             family=Gamma(link=log) )  
summary(glm1)
```

Interpreting the output

```
Call:
lm(formula = y ~ year)

Residuals:
    Min       1Q   Median       3Q      Max
-2.25429 -0.45643 -0.02429  0.75571  2.11786

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -1774.9657    674.7740  -2.630   0.0234 *
year           0.8879     0.3369   2.635   0.0232 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.211 on 11 degrees of freedom
Multiple R-squared:  0.387,    Adjusted R-squared:  0.3313
F-statistic: 6.945 on 1 and 11 DF,  p-value: 0.02318
```

Interpreting the output

```
Call:
glm(formula = response1 ~ 1, family = Gamma(link = log), weights = weight1)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-2.39018  -1.06068  -0.45670   0.05158   1.18756

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   0.3830     0.3646   1.05   0.314

(Dispersion parameter for Gamma family taken to be 0.7313267)

    Null deviance: 16.279  on 12  degrees of freedom
Residual deviance: 16.279  on 12  degrees of freedom
AIC: 15.869

Number of Fisher Scoring iterations: 8
```

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AIC: 15.869

Number of Fisher Scoring iterations: 8

Exercise in pairs: Calculate a 95% CI for the residual variance.

Modelling the residual variance

```
glm2 <- glm( response1 ~ year, weights = weight1 ,  
             family=Gamma(link=log) )  
summary(glm2)
```

Output

```
Call:
glm(formula = response1 ~ year, family = Gamma(link = log), weights = weight1)
```

Deviance Residuals:

Min	1Q	Median	3Q	Max
-2.4203	-0.9836	-0.5411	0.1114	1.0389

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	384.3795	744.4281	0.516	0.616
year	-0.1917	0.3717	-0.516	0.616

(Dispersion parameter for Gamma family taken to be 0.6987414)

Null deviance: 16.279 on 12 degrees of freedom
Residual deviance: 16.131 on 11 degrees of freedom
AIC: 17.801

Number of Fisher Scoring iterations: 13